1 Tests for Convergence

In general, it is difficult to find the exact sum of a series. We were able to accomplish this for geometric and telescoping series, because in each of those cases we could find a simple formula for the nth partial sum s_n. However, it usually isn't easy to discover such a formula. Therefore, we seek to develop tests that will enable us to determine whether a series is convergent or divergent without explicitly finding its sum. In some cases, these methods will also enable us to find good estimates of the sum.

2 Integral Test

Last time, we proved the Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by looking at the sequence of partial sums $\{s_n\}$ and showing that $s_{2^n} > 1 + \frac{n}{2}$ for all $n \geq 2$. We now give an alternative argument that the Harmonic Series diverges, which will be applicable to many other series as well.

1. Interpret the nth partial sum $\sum_{i=1}^{n} \frac{1}{i}$ of the Harmonic Series as a left-Riemann sum $\sum_{i=1}^{n} f(x_i^*)\Delta x$ for some integral $\int_a^b f(x)dx$. Identify $f(x)$, $[a,b]$, x_i^*, and Δx, write the integral $\int_a^b f(x)dx$ corresponding to s_n, and draw the graph of $f(x)$ and the rectangles of the Riemann sum for $n = 5$.

2. How does s_n compare to the integral it is approximating?

The method we just used to show that the Harmonic Series diverges can be applied to many other series.

Theorem 2.1 (The Integral Test)**.** *Suppose f is a continuous, positive, decreasing function for $x \geq 1$, and let $a_n = f(n)$ for $n = 1, 2, 3, \ldots$. Then*

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x)dx$$

either both converge or both diverge.

Before proving the integral test, we consider some examples.

4. Use the Integral Test to determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges or diverges.

5. For which values of p does the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

The series in the previous example is called the *p-series*. It will be important later, so we summarize the results of the previous example for future reference:

The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

6. Determine whether each of the following p-series converge or diverge.

   (a) $\sum_{n=1}^{\infty} \frac{1}{n^3}$

   (b) $\sum_{n=1}^{\infty} \frac{1}{n^{1/3}}$

Remark 2.2. One should *not* infer from the Integral Test that the sum of the series is equal to the value of the integral. For example, it was proved by Euler in 1736 that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, whereas $\int_1^{\infty} \frac{dx}{x^2} = 1$. Therefore, in general, $\sum_{n=1}^{\infty} a_n \neq \int_1^{\infty} f(x)dx$.

7. Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln(n)}{n}$ converges or diverges.

3 Proof of the Integral Test

Theorem 2.1 (The Integral Test) Suppose f is a continuous, positive, decreasing function for $x \geq 1$, and let $a_n = f(n)$ for $n = 1, 2, 3, \ldots$. Then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x)dx$$

either both converge or both diverge.

Proof. 8. Show that the sequence $s_n = \sum_{i=1}^{n} a_i$ of partial sums is increasing.

9. Show that for $n \geq 1$ we have

$$\sum_{i=2}^{n} a_i \leq \int_1^n f(x)dx \leq \sum_{i=1}^{n-1} a_i. \tag{3.1}$$

10. Use the first inequality in Equation (3.1) to show that if $\int_1^\infty f(x)dx$ converges, then the sequence $s_n = \sum_{i=1}^n a_i$ is bounded above. Conclude that the series $\sum_{i=1}^\infty a_i$ converges if $\int_1^\infty f(x)dx$ does.

11. What is the only way that $\int_1^\infty f(x)dx$ can diverge?

12. Use result of the previous question together with the second inequality in Equation (3.1) to show that $\sum_{i=1}^\infty a_i$ diverges if $\int_1^\infty f(x)dx$ does.

□